INCH-POUND

MIL-C-24643/3C
14 March 1994
SUPERSEDING
MIL-C-24643/3B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPES LSSHOF, LSDHOF, LSTHOF AND LSFHOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated, (see table I for size)
* Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber (see table I for wall thickness). Standard identification code applied by method 3 or 4. Note a colored tape may be used on size 23 and larger.
Fourth - Reinforcement on type LSSHOF, sizes 23 and larger.
Fifth - The required number of conductors cabled together with a left hand lay (see table I for length of lay). Fillers shall be employed on multi-conductor types to obtain a firm well rounded assembly.
Sixth - Binder tape applied helically with overlap on multi-conductor cables.
* Seventh - Cross-linked polyolefin jacket.

NOTE: The initial letter of the cable designation denotes the number of conductors. Thus "S" denotes a single conductor cable; "D" a two conductor cable; "T" a three conductor cable; and "F" a four conductor cable.

AMSC N/A FSC 6145

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TABLE I. Details.

Military part no. 24643/3	Type and size	Conductors size		Insulation thickness min avg (in)	Lay of conductor max (in)	Jacket wall thickness min avg (in)	Overall Diameter		Roller dia for bending endurance (max) (in)	Conductores 1000 ft (max ohms)	Insulation resistance (megohm)
		Navy std	AWG				min (in)	max (in)			
-01UN	LSSHOF-3		16 (CLASS M)	0.031		0.030	0.195	0.210		4.28	300
-02UN	LSSHOF-23		7 (CLASS K)	.040		.040	.440	.460		.546	300
-03UN	LSSHOF-60	60 (304)		.050		.050	.570	.600		.187	250
-04UN	LSSHOF-150	150 (760)		.070		.050	.830	.870		.0747	200
-05UN	LSSHOF-200	200 (988)		.070		.050	.940	.980		.0575	200
-06UN	LSSHOF-250	250 (1254)		.070		.070	1.035	1.085		.0453	150
-07UN	LSSHOF-500		500MCM (CLASS K)	.090		.070	1.380	1.450		.0234	100
-08UN	LSSHOF-650		650MCM (CLASS G)	.100		.070	1.540	1.610		.0174	100
-09UN	LSSHOF-800	800 (4033)		.100		.070	1.600	1.670		.0141	100
-10UN	LSDHOF-3		16 (CLASS K)	.031	2	.040	0.405	.425	1.3	4.36	300
-11UN	LSDHOF-4		14 (CLASS K)	.031	2	.040	.440	.460	1.4	2.73	250
-12UN	LSDHOF-6		12 (CLASS K)	.031	2-1/2	.040	.490	.510	1.5	1.72	200
-13UN	LSDHOF-9	9(90)		.031	3	.050	.540	.570	1.7	1.24	200
-14UN	LSDHOF-14	14 (140)		.040	4	.050	.675	.705	2.1	.802	200
-15UN	LSDHOF-23		7(CLASS K)	.040	5-1/2	.050	.820	.860	3.0	0.560	150
-16UN	LSDHOF-30		5(CLASS K)	.050	7-1/2	.050	.920	.960	3.5	.351	150

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TABLE I. Details - Continued.

Military part no. 24643/3	Type and size	Conductors size		Insulation thickness min avg (in)	Lay of conductor max (in)	Jacket wall thickness min avg (in)	Overall Diameter		Roller dia for bending endurance (max) (in)	Conductor resistance 1000 ft (max ohms)	Insulation resistance (meg-ohm)
		Navy std	AWG				min (in)	max (in)			
-17UN	LSDHOF-83	83 (418)		.070	16	.070	1.390	1.450	7.0	.139	150
-18UN	LSDHOF-250	250 (1254)		.070	24	.100	2.000	2.100		.0462	100
-19UN	LSDHOF-400	400 (2052)		.090	30	.100	2.400	2.500		.0283	100
-20UN	LSTHOF-3		16(CLASS K)	0.031	2	.040	0.430	0.450	1.4	4.36	300
-21UN	LSTHOF-4		14(CLASS K)	.031	2-1/2	.040	.460	.480	1.4	2.73	250
-22UN	LSTHOF-6		12(CLASS K)	.031	3	.050	.520	.550	1.7	1.72	200
-23UN	LSTHOF-9	9(90)		.031	3-1/2	.050	.570	.600	1.8	1.24	200
-24UN	LSTHOF-14	14 (140)		.040	4-1/2	.050	.720	.750	2.3	.802	200
-25UN	LSTHOF-23		7(CLASS K)	.040	6	.050	.860	.900	3.5	.560	150
-26UN	LSTHOF-42	42 (209)		.070	12	.070	1.200	1.250	4.5	.277	150
-27UN	LSTHOF-150	150 (760)		.070	18	.100	1.740	1.820		.0762	125
-28UN	LSTHOF-250	250 (1254)		.070	24	.100	2.140	2.240		.0462	100
-29UN	LSTHOF-400	400 (2052)		.090	30	.100	2.680	2.800		.0283	100
-30UN	LSTHOF-500		500MCM (CLASS K)	.090	33	.100	2.920	3.100		.0230	100
-31UN	LSTHOF-600		600MCM (CLASS G)	.100	36	.100	2.980	3.150		.0200	100
-32UN	LSFHOF-3		16 (CLASS K)	.031	2-1/2	.040	.460	.480	1.4	4.36	300
-33UN	LSFHOF-4		14 (CLASS K)	.031	3	.040	.520	.550	1.6	2.73	250
-34UN	LSFHOF-9	9(90)		.031	4	.050	.630	.660	1.8	1.24	200
-35UN	LSFHOF-42	42 (209)		.070	15	.070	1.300	1.380	4.1	.277	150
-36UN	LSFHOF-60	60 (304)		.070	18	.070	1.430	1.510	4.5	.191	150
-37UN	LSFHOF-133	133 (684)		.070	20	.100	1.920	2.000		.0847	125

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EXAMINATION AND TESTS:	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	
Sizes 3 to 9, inclusive	2000
Sizes 14 to 800, inclusive	2500
Insulation resistance - megohms/1000 feet, minimum	(see table I)
Conductor continuity	No failure
Group A	
Visual and dimensional	No failure
* Group B	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (when tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Twisting and bending endurance - (see note 1)	
At 75°C, cycles, minimum	6000
At minus 30 ± 2°C, cycles, minimum	2000
Group C	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60

NOTE: 1. Tests to be performed on all sizes of LSDHOF, LSTHOF and LSFHOF; excepting LSSHOF-250, LSDHOF-400, LSTHOF-150, LSTHOF-250, LSTHOF-400, LSTHOF-500, LSTHOF-600, LSFHOF-133. Test not required for LSSHOF. (See table I for roller sizes).

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EXAMINATION AND TESTS (continued):

	Requirements
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (jacket) - cycles, minimum	125

* Group D

Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSSHOF -	23	2000
LSSHOF -	60 thru 250	1000
LSSHOF -	500 thru 800	500
LSDHOF -	6	2000
LSDHOF -	9 and 14	1500
LSDHOF -	23 thru 83	1000
LSDHOF -	250 and 400	500
LSTHOF -	6	2000

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UNIT ORDERING LENGTHS (continued):

LSTHOF -	9 and 14	1500
LSTHOF -	23 thru 150	1000
LSTHOF -	250 and 600	500
LSFHOF -	4	2000
LSFHOF -	9	1500
LSFHOF -	42 and 60	1000
LSFHOF -	133	500

Note: (LSSHOF-3, LSDHOF-3, LSDHOF-4, LSTHOF-3, LSTHOF-4, LSFHOF-3) packed: 500 feet (nominal)

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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